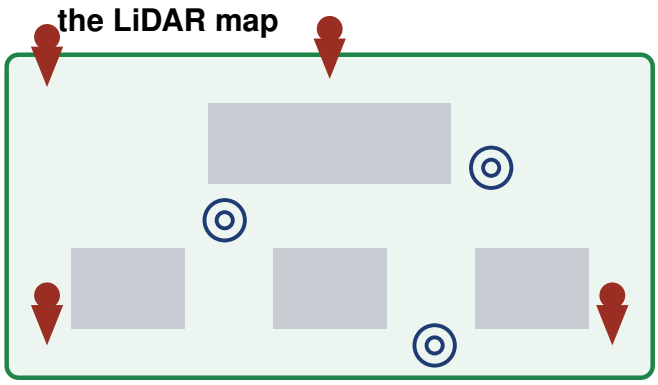
Control, Error Budgets and Accuracy Standards

BIG IDEA

Control points pin the map to the real world, and secret checkpoints test it.



control point: pins the map in place

checkpoint: kept secret, used only to test the map

THE IDEA

You can draw a perfect map of your bedroom, with every toy in the right place compared with every other toy. But if you don't know where your house is on the street, nobody else can use it to find your room. SLAM makes the bedroom map. **Control points**, measured very carefully by surveyors, pin that map onto the real world.

REMEMBER

- ✓ A map can look perfect and still be in the wrong place.
- ✓ **Checkpoints** are kept secret from the computer and used only to test the map.
- ✓ A formal test uses at least 30 checkpoints, spread over the whole project.

TRY THIS

Draw a map of your room. Pick three spots and measure them with a tape measure. Does your map agree with the tape?